National University of computer and emerging sciences

Artificial Intelligence Lab (AI-2002)
Semester: Spring 2026
Section: BCS-6A & BCS-6F
Course Instructor: Mr. Abdullah Shaikh

LAB # 03 - TASKS

Agents and Environment

Submission Instructions:

- Perform all tasks in class only. Late submissions will not be considered.
- Submit a .ipynb file with the filename formatted as:
  Lab 03 - Your Name - Roll Number
- Upload the file on GCR.

TASK # 01

You are designing a system for an office building that automatically controls the lights in each room. The lights should turn on or off based on the presence of people.

Requirements:

1. Each room has a motion sensor that detects whether someone is present.
2. If the room is occupied, the light should be turned on immediately.
3. If the room becomes empty, the light should be turned off immediately.
4. The system should operate continuously, reacting to the current state of each room.
5. The system should not store any history about previous occupancy; it should act based on what the sensor currently detects.

TASK # 02

You are designing a robot that manages inventory in a warehouse. The robot must pick items from storage shelves and deliver them to packing stations.

Requirements:

1. The warehouse is large with multiple aisles and shelves.
2. The robot has sensors that detect:
   - Current location (aisle and shelf)
   - Whether the current shelf has the required item
3. The robot does not always see all shelves at once, so it must remember which shelves it has already checked.
4. If an item is missing on a shelf, the robot should remember this and avoid checking the same empty shelf repeatedly.
5. The robot must plan a route to efficiently pick all required items.
6. The robot must continue working until all items on the pick list are collected.

Collect all items on the pick list efficiently, using past information about shelves to avoid unnecessary checks or movements.

TASK # 03

You are designing an autonomous drone that delivers packages in a city. The drone must choose its route and speed to balance delivery speed, battery usage, and safety.

Requirements:

1. The drone can sense:
    - Current battery level
    - Weather conditions (wind, rain, etc.)
    - Distance to delivery destination
    - Obstacles in the flight path
2. Possible actions:
    - Fly faster
    - Fly slower
    - Take an alternative route
3. Each action affects different factors:
    - Flying faster → delivers sooner but consumes more battery and may be riskier in bad weather
    - Flying slower → safer and more battery-efficient, but takes longer
    - Alternative route → may avoid obstacles but could increase distance or time
4. The drone must evaluate trade-offs between delivery time, battery consumption, and safety to maximize overall utility.
5. The drone should adapt in real-time to changing weather, battery level, or obstacles.

Goal:

Deliver packages as quickly as possible, while ensuring enough battery to complete the trip safely and avoiding dangerous conditions, by selecting the action that maximizes combined utility.

Hints for Possible Values

Table 1: TASK # 03

Parameter	Sample Values	Notes
Battery	0–100	percentage of remaining battery
Weather	Clear, Windy, Rainy, Stormy	affects safety and risk
Distance (km)	1–20	distance to destination
Obstacles	0–5	number of obstacles in flight path
Actions	Fly Faster, Fly Slower, Alternative Route	choose one maximizing utility

THE END!